TradeIQ (StockSense AI)

System Architecture, Quantitative Engine Pipeline, Technology Stack & Market Differentiation

1. Executive Summary

TradeIQ (StockSense AI) is an institutional-grade algorithmic equity analysis and tactical research terminal. It fuses real-time market data, deterministic technical indicators, multi-factor fundamental valuation, financial NLP sentiment scoring, and quantitative risk modeling (Sharpe, Beta, VaR, CVaR) with dual-layer LLM reasoning (LangChain + ChatGroq). The system replaces speculative opinions with mathematical evidence, generating Wall Street caliber research memorandums in sub-second latency.

2. System Architecture & End-to-End Pipeline

The application is structured into decoupled, resilient tiers designed for zero-latency execution, deterministic fallbacks, and strict security compliance:

Tier	Core Components & Technologies	Functional Responsibility
1. Presentation Layer	Vanilla HTML5, Modern CSS Design Tokens, Modular JS, ApexCharts 7.1.0, Inter Font	Institutional terminal UI, interactive candlestick/area charting, Microsoft-style 2-step auth card, live parameter tweaking, and client-side PDF export.
2. Gateway & API Layer	FastAPI 0.136, Uvicorn ASGI, Pydantic V2 Schemas, SlowAPI Rate Limiter, CORS Middleware	Validates ticker symbols (`^[A-Z0-9.-]{1,10}$`), enforces rate-limits (20 req/min), handles JWT session bearer tokens, and routes endpoints.
3. Quantitative Analysis Core	Pandas 3.0, NumPy 2.2, native indicator routines (SMA 20/50/200, RSI-14, MACD, Bollinger Bands, Volatility)	Ingests OHLCV price action from Yahoo Finance, strips numba dependencies for 100% platform portability, and derives multi-timeframe mathematical signals.
4. Sentiment & NLP Subsystem	VADER Sentiment, YFinance News API, Google News RSS Fallback, regex sanitization	Aggregates breaking company news headlines, calculates compound polarity scores (-1.0 to +1.0), and generates numerical sentiment evidence.
5. Institutional Risk & Backtest	SciPy Statistical Distribution, Vectorized Pandas simulation, VaR (95%), CVaR, Sharpe, Beta	Executes 1-year historical signal backtests (RSI + SMA50 crosses) to calculate strategy win rate and evaluates tail-risk exposure.
6. LLM Reasoning Layer	LangChain Groq (Llama 3.3 70B Versatile), ChatPromptTemplate, JsonOutputParser, Rule Fallback	Synthesizes multi-factor data points into structured investment memorandums. Falls back to a deterministic mathematical heuristic if LLM limits trigger.

7. Persistence & Auth	SQLAlchemy 2.0 ORM, SQLite / PostgreSQL, Passlib (Bcrypt), Python-Jose (JWT)	Secure user authentication, encrypted credential storage, watchlists, risk tolerance profiles, and portfolio preference persistence.

3. Detailed Data Pipeline Flow

Step 1 — Ingestion: When an equity ticker is selected, `services.stock_data` fetches 1-year historical daily OHLCV and fundamental balance sheet info.
Step 2 — Feature Engineering: `services.technical` executes rolling window convolutions to establish trend momentum, volatility, and mean-reversion bands.
Step 3 — Fundamental Valuation: `services.fundamental` analyzes P/E, Diluted EPS, YoY Revenue Growth, Profit Margins, and Debt-to-Equity relative to baseline thresholds.
Step 4 — News Sentiment Extraction: `services.sentiment` queries real-time financial headlines, executing VADER lexical analysis with RSS failover.
Step 5 — Strategy Backtesting: `services.quant_backtest` runs an historical trade signal simulation over 252 trading days, tracking accuracy and non-double-counted signals.
Step 6 — LLM Synthesis or Algorithmic Fallback: LangChain constructs a structured JSON prompt passed to Groq Llama 3.3. If Groq is unavailable, the deterministic weighted scoring engine (45% Tech + 35% Fund + 20% Sent) delivers an uncompromised report.

4. Comprehensive Technology Stack Taxonomy

Category	Technology / Library	Version / Role
Backend Framework	FastAPI, Starlette, Uvicorn ASGI	High-concurrency async REST API, auto-docs
AI & LLM Orchestration	LangChain, LangChain-Groq, Llama-3.3-70B	Structured output parsing, prompt templating, reasoning
Data Science & Quant	Pandas 3.0, NumPy 2.2, SciPy 1.17	Time-series convolutions, VaR, CVaR, annualized metrics
Market Data Feed	yfinance, curl_cffi, Feedparser	Real-time tick data, balance sheet info, RSS news feed
NLP & Sentiment	vaderSentiment, BeautifulSoup4	Lexical sentiment polarity scoring and headline parsing
Database & ORM	SQLAlchemy 2.0, SQLite, PostgreSQL Driver	Declarative relational models, watchlist management
Security & Auth	Python-Jose (JWT), Passlib (Bcrypt), SlowAPI	Stateless token auth, rate limiting (20/min), password hashing
Frontend Terminal	HTML5, Modern Vanilla CSS, ES6 JavaScript	Light/Dark modes, zero dependencies, Inter font
Data Visualization	ApexCharts 7.1.0 (Lazy Loaded)	Interactive financial area, candlestick, and backtest graphs
Testing & Quality	Pytest 9.1, Ruff 0.17, AnyIO	39 automated unit/integration tests, zero lint warnings
Deployment & DevOps	Docker Multi-Stage, GitHub Actions CI, Render	Continuous deployment, automated containerization

5. What Makes TradeIQ Stand Out in the Market

Retail investors and professional analysts face a bifurcated market: complex, prohibitively expensive terminals (Bloomberg at $24,000/yr) versus superficial retail trading apps (Robinhood, Yahoo Finance) that lack rigorous risk modeling. TradeIQ bridges this gap through four key pillars of differentiation:

Competitive Vector	Traditional Retail Platforms (Yahoo/Robinhood)	TradeIQ (StockSense AI) Institutional Advantage
1. Multi-Factor Synthesis	Siloed data: users must manually cross-reference price charts with news tabs and P/E ratios.	**Unified Composite Quant Score (0-100)**: Evaluates Technical (45%), Fundamental (35%), and Sentiment (20%) in one coherent calculation.

2. AI Grounding & Anti-Hallucination	Generic AI chatbots hallucinate stale prices, speculative rumors, and unsupported claims.	**Zero-Hallucination Evidence Constraint**: LLM can only cite computed mathematical inputs. If an indicator is missing, confidence is penalized automatically.
3. Strategy Backtest Validation	Zero historical accountability. Users act on blind faith or promotional commentary.	**Built-in 1-Year Historical Backtest Engine**: Proves actual win rates and trade counts before suggesting position entries.
4. Institutional Risk Metrics	Only basic volatility or 52-week high/low metrics are displayed.	**Hedge-Fund Grade Risk Analytics**: Quantifies Value at Risk (VaR 95%), Expected Shortfall (CVaR), Sharpe Ratio, and S&P; 500 Beta.
5. Deterministic Failover	Complete outage if third-party LLM APIs suffer downtime or rate limits.	**Dual-Model Failover + Algorithmic Rule Engine**: 100% uptime guaranteed even during AI provider interruptions.
6. User Experience & Accessibility	Cluttered ad banners, slow heavy page loads, poor mobile responsiveness.	**Sleek Terminal & Microsoft Account UX**: Underline inputs, multi-step card flow, password manager integration, WCAG AA 4.5:1 contrast, and sub-second load times.

6. Architectural Topology & Signal Synthesis Flow

The following diagram illustrates the linear and parallel signal pathways within the QuantCore engine:

[CLIENT BROWSER / TERMINAL] **TradeIQ Minimal UI • ApexCharts 7.1.0 • Inter UI • Microsoft-style Auth Flow**
↓ *HTTPS REST / JWT Bearer Tokens / 20 req/min Rate Limiting*
[FASTAPI GATEWAY & ROUTER LAYER] **`routers/api.py` • `routers/auth.py` • Pydantic Schema Validation**
↓ *Parallel Asynchronous Micro-Pipelines*
[QUANTCORE MULTI-FACTOR ENGINE] • **Technical Engine:** SMA 20/50/200, RSI-14, MACD, Bollinger Bands, Volatility • **Fundamental Engine:** Trailing/Forward P/E, Diluted EPS, YoY Revenue Growth, Profit Margins, Debt/Equity • **Sentiment Engine:** Dual-source Headline Aggregation + VADER Lexicon Compound Polarity • **Risk Engine:** Value at Risk (VaR 95%), Expected Shortfall (CVaR), Sharpe Ratio, Beta • **Backtest Engine:** 252-day Historical Cross Simulation (Strategy Accuracy & Win Rate)
↓ *Normalized Evidence Vector*
[REASONING & SYNTHESIS TIER] **Primary:** LangChain + ChatGroq (Llama-3.3-70B Versatile, temp=0.0, JsonOutputParser) **Secondary:** Secondary Groq Model Candidate Failover **Tertiary Fallback:** Deterministic Algorithmic Quant Heuristic (Zero-Downtime Guarantee)
↓ *Standardized Structured Output*
[EXECUTIVE QUANT MEMORANDUM & DECISION BADGE] **Rating (BUY / HOLD / SELL) • Confidence (0-100%) • Portfolio Allocation • Technical Evidence**

7. Enterprise Security, Reliability & Quality Assurance

• **Authentication & Password Hardening:** Passwords hashed using industry-standard Bcrypt (`passlib`). Full support for password manager autofill pairing across multi-step auth forms.
• **Session Governance:** Dual persistence model — `localStorage` for 'Keep Me Signed In' sessions; ephemeral `sessionStorage` for shared devices.
• **API Defense:** In-memory rate limiting via SlowAPI (20 req/min per IP) to mitigate scrapers and DDoS attacks.
• **Continuous Integration:** GitHub Actions workflow executing Ruff linting, Docker build testing, and 39 automated Pytest validation cases.
• **Accessibility & Performance:** WCAG AA compliant contrast ratios (`#556277` text muted), self-hosted Inter font, inline SVGs, zero render-blocking scripts, and sub-50ms DOM painting.

8. Conclusion & Strategic Value Proposition

TradeIQ successfully converts raw, overwhelming financial data into disciplined, actionable intelligence. By enforcing strict mathematical grounding on advanced language models, the platform gives traders, quants, and wealth managers an edge once reserved for tier-one hedge funds.